



MIT

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1. With detail case study on Demings and Ishikawa's contributions to quality management

1. Introduction

Quality management is the process of ensuring that products and services consistently meet customer requirements and organizational standards. Two of the most influential thinkers in the field of quality management are **Dr. W. Edwards Deming** and **Dr. Kaoru Ishikawa**. Their ideas played an important role in the development of modern **Total Quality Management (TQM)**.

Deming emphasized **continuous improvement, statistical quality control, management responsibility, and reduction of variation**, while Ishikawa emphasized **employee participation, quality circles, root-cause analysis, and practical quality-control tools**.

2. W. Edwards Deming's Contributions to Quality Management

2.1 Background

Dr. W. Edwards Deming was an American statistician and quality management expert. He made significant contributions to industrial quality management and became particularly influential in Japan after World War II.

He taught Japanese engineers and managers how statistical methods and systematic management practices could be used to improve product quality and productivity.

2.2 Major Contributions of Deming

A. Deming's 14 Principles of Management

Deming proposed **14 principles** to help organizations achieve continuous improvement. Some important principles include:

1. Create a constant purpose for improvement.
2. Adopt a new philosophy of quality.
3. Reduce dependence on mass inspection.
4. Improve production and service continuously.
5. Provide proper training to employees.
6. Develop effective leadership.
7. Remove fear from the workplace.
8. Break down barriers between departments.
9. Encourage employee participation.
10. Continuously improve processes and systems.

These principles emphasize that quality is the responsibility of **management and the entire organization**, not only the inspection department.

B. PDCA Cycle

Deming promoted the **PDCA cycle** as a method for continuous improvement.

- **Plan:** Identify a problem and plan an improvement.
- **Do:** Implement the planned solution.
- **Check:** Measure and evaluate the results.
- **Act:** Standardize successful improvements and continue the cycle.

Thus:

Plan → Do → Check → Act → Continuous Improvement

C. Statistical Quality Control

Deming emphasized the use of **statistical techniques** to identify and control variations in production processes

Instead of simply inspecting finished products, organizations should monitor the process itself and identify problems before they result in defective products.

D. Reduction of Variation

Deming considered excessive variation in processes to be a major cause of poor quality. He encouraged organizations to identify the causes of variation and make processes more stable and predictable.

E. Management Responsibility

According to Deming, management plays a major role in creating quality. Management should provide proper training, resources, leadership, and an environment where employees can contribute to improvement.

3. Case Study: Deming and Japanese Industries

After World War II, Japanese industries needed to rebuild their manufacturing sector and improve the quality of their products.

Deming conducted lectures and training programs for Japanese engineers and managers. He introduced concepts such as **statistical process control, continuous improvement, and management responsibility for quality**.

Implementation

Japanese companies began to:

- Use statistical techniques to monitor production.
- Train employees in quality-control methods.
- Reduce variation in manufacturing processes.
- Improve processes continuously.
- Focus on customer requirements.
- Involve management and employees in quality improvement.

Outcome

These practices contributed to significant improvements in the **quality, reliability, productivity, and international competitiveness** of Japanese products.

Lesson from the Case

The major lesson from Deming's approach is:

“Quality should be built into the process rather than achieved only through inspection of the final product.”

4. Kaoru Ishikawa's Contributions to Quality Management

4.1 Background

Dr. Kaoru Ishikawa was a prominent Japanese quality management expert. He played an important role in developing quality-control practices in Japan.

His approach focused on **employee involvement, teamwork, root-cause analysis, and company-wide quality management.**

4.2 Major Contributions of Ishikawa

A. Cause-and-Effect Diagram

Ishikawa developed the **Cause-and-Effect Diagram**, commonly known as the **Ishikawa Diagram** or **Fishbone Diagram**.

It is used to identify and organize the possible causes of a particular problem.

The commonly used categories are known as the **6 Ms**:

1. **Man** – People or workers
2. **Machine** – Equipment and machinery
3. **Method** – Procedures and processes
4. **Material** – Raw materials
5. **Measurement** – Measurement systems
6. **Mother Nature/Environment** – Working environment

For example, if a factory experiences a high number of defective products, the fishbone diagram can help identify whether the problem is caused by workers, machines, materials, methods, or other factors.

B. Quality Circles

Ishikawa strongly promoted the concept of **Quality Circles**.

A quality circle is a small group of employees who regularly meet to identify workplace problems, analyze their causes, and suggest solutions.

Quality circles encourage:

- Employee participation
- Teamwork
- Problem-solving
- Better communication
- Continuous improvement

C. Company-Wide Quality Control

Ishikawa believed that quality should not be restricted to the production or inspection department.

Every department should contribute to quality, including:

- Production
- Engineering
- Marketing
- Sales
- Purchasing

- Management
- Employees

Therefore, quality becomes a **company-wide responsibility**.

D. Seven Basic Quality Tools

Ishikawa helped popularize the use of simple quality-control tools that could be understood and used by employees at different levels of an organization.

The seven basic quality tools include:

1. Cause-and-effect diagram
2. Check sheet
3. Control chart
4. Histogram
5. Pareto chart
6. Scatter diagram
7. Stratification/Flowchart

These tools help organizations collect data, identify problems, analyze causes, and monitor improvements.

5. Case Study: Ishikawa and Quality Circles

Consider a manufacturing company where **10% of the products are defective**.

The company forms a quality circle consisting of workers, supervisors, and engineers.

Step 1: Collect Data

The team uses a **check sheet** to record different types of defects.

Step 2: Identify Major Problems

A **Pareto chart** is used to identify which defect occurs most frequently.

Step 3: Identify Possible Causes

The team uses an **Ishikawa/Fishbone diagram** to identify possible causes related to:

- Workers
- Machines
- Materials
- Methods
- Measurement
- Environment

Step 4: Find the Root Cause

The team analyzes the production process and identifies the major cause of the defects.

Step 5: Implement Corrective Action

The company changes the relevant process, provides training, or improves the equipment.

Step 6: Monitor the Results

A **control chart** or other quality tool is used to monitor the process after improvement.

Outcome

This approach can help the organization:

- Reduce defects
- Reduce waste
- Improve productivity
- Increase employee involvement
- Improve customer satisfaction
- Promote continuous improvement

6. Comparison of Deming and Ishikawa

Basis	W. Edwards Deming	Kaoru Ishikawa
Main focus	Continuous improvement and management	Employee participation and problem-solving
Major contribution	14 Principles and PDCA approach	Fishbone/Cause-and-Effect Diagram
Quality approach	Statistical and systematic	Practical and participative
Employee role	Employees should be trained and involved	Employees actively participate through quality circles
Major concept	Reduction of variation	Root-cause analysis
Quality responsibility	Strong emphasis on management responsibility	Company-wide responsibility
Important tools/concepts	PDCA, statistical quality control	Quality circles, 7 basic quality tools
Overall objective	Continuous improvement of systems and processes	Identifying and solving quality problems

7. Overall Impact on Quality Management

The contributions of Deming and Ishikawa helped establish several important principles of modern quality management:

- Quality is everyone's responsibility.
- Customer satisfaction is an important objective.
- Quality should be built into processes.
- Problems should be prevented rather than simply detected.
- Employees should participate in quality improvement.
- Decisions should be supported by data.
- Root causes should be identified before taking corrective action.
- Organizations should continuously improve their processes.

8. Conclusion

W. Edwards Deming and Kaoru Ishikawa made significant contributions to the development of modern quality management. Deming emphasized **management responsibility, statistical quality control, reduction of variation, and continuous improvement through the PDCA approach.** Ishikawa emphasized **employee participation, quality circles, root-cause analysis, and practical quality-control tools such as the Fishbone Diagram.**

Their ideas demonstrated that quality cannot be achieved only through final inspection. Instead, it requires **continuous improvement, effective management, employee involvement, teamwork, and systematic problem-solving**. Their contributions continue to influence **Total Quality Management (TQM)** and quality practices in organizations around the world.